

Numerical Analysis First-Year Exam, January 2013

Answer any 4 questions.

1. Suppose that $A \in \mathbb{C}^{n \times n}$ is Hermitian with λ_1 and λ_n the smallest and largest eigenvalues of A respectively. Show that

$$\lambda_1 \|x\|^2 \leq x^* A x \leq \lambda_n \|x\|^2$$

for all $x \in \mathbb{C}^n$.

2. This problem has three parts.

- (a) Suppose p and $q \in \mathbb{R}$ with p and q positive and $p^{-1} + q^{-1} = 1$. Show that for any matrix $A \in \mathbb{C}^{m \times n}$, we have $\|A\|_p = \|A^*\|_q$, where A^* is the conjugate transpose of A .

- (b) Prove that

$$\|A\|_2^2 \leq \|A\|_p \|A\|_q$$

for any $A \in \mathbb{C}^{m \times n}$ and any positive p and $q \in \mathbb{R}$ with $p^{-1} + q^{-1} = 1$.

- (c) Prove that for any $p \geq 1$ and any diagonal matrix $D \in \mathbb{C}^{n \times n}$, we have

$$\|D\|_p = \max\{|d_{ii}| : 1 \leq i \leq n\}.$$

3. This problem has two parts.

- (a) Show that the eigenvalues of a projector are either 0 or 1.

- (b) Show that a projector P is orthogonal if and only if $P = P^*$.

4. Show that the element of largest magnitude in a Hermitian, positive definite matrix lies on the diagonal.

5. Suppose that $A \in \mathbb{C}^{n \times n}$ is an invertible matrix, $u \in \mathbb{C}^n$, and $v \in \mathbb{C}^n$.

- (a) Show that

$$\det(A - uv^*) = (1 - v^* A^{-1} u) \det A.$$

- (b) Suppose that A is a real diagonal matrix and $u = v$ with $u_i \neq 0$ for each i . Show that the eigenvalues of $A - uu^*$ are the roots of

$$\sum_{i=1}^n \frac{|u_i|^2}{a_{ii} - \lambda} = 1.$$